

INTERNSPARK PORTFOLIO REPORT: TASK 1

Candidate Information

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Domain: Machine Learning Intern

Project Name: Iris Flower Species Classification

1. Objective & Business Use Case

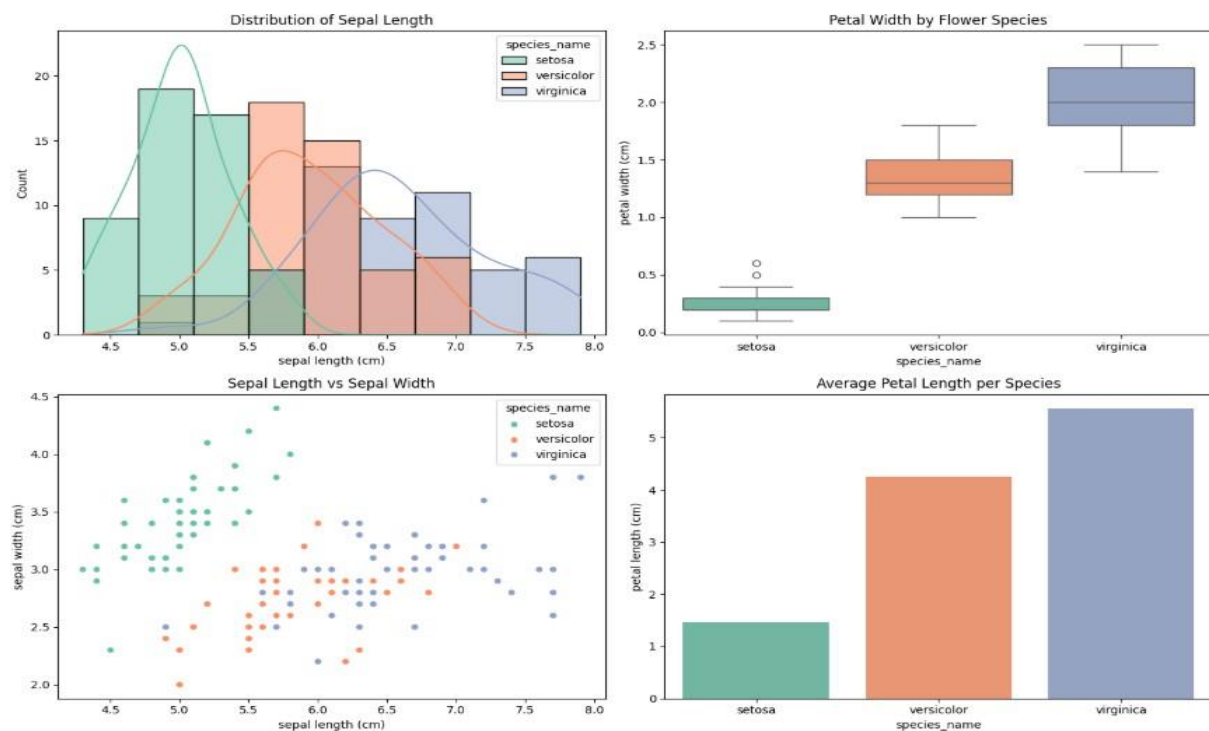
The purpose of this project is to implement an end-to-end supervised machine learning pipeline to classify Iris flower species (Setosa, Versicolor, Virginica) based on morphometric features. This acts as a foundation for implementing deployment-ready predictive modeling workflows.

2. Exploratory Data Analysis (EDA) Summary

Through 4 distinct visual representations, the data trends showed:

Clear linear separation of the Setosa species based on petal length and width.

Slight data overlap between Versicolor and Virginica, requiring a clear decision boundary.



3. Methodology & Performance

Preprocessing: Data was normalized using `StandardScaler` to handle multi-unit scales.

Model Selection: A baseline Logistic Regression model was trained on an 80/20 train-test split.

Evaluation Metrics: The model achieved an accuracy of [Insert your cell output accuracy, e.g., 100% or 96.67%] on unseen test configurations.

